FYP MEETING LOG

Co-Supervisor Meeting 1 of 2

Agent-Based Game Analysis in Unity Based Upon Curiosity and RND Exploration Using PPO and SAC
Student: Raza Ullah Khan (NUM-BSCS23-28) | Supervisor: Dr. Malik M Ali Shahid | Co-Supervisor: Mr. Abdul Rafay | Industrial Partner: Inspired Square

Meeting No.	03 (Co-Supervisor Meeting 1)
Meeting With	Co-Supervisor - Mr. Abdul Rafay
Date	Monday, 19 October 2026
Time	11:00 AM - 11:20 AM
Duration	20 minutes
Mode / Venue	Co-Supervisor's office, Namal University Mianwali
Attendees	Raza Ullah Khan (Student); Mr. Abdul Rafay (Co-Supervisor)
Milestone	FN-01 (Month 1, Weeks 1-2): Project Initialization, Development Environment & Version Control

1. Purpose of the Meeting

A technical meeting to decide the exact software versions and set up the development environment and version control in the right way from day one.

2. Agenda

1. Which Unity and ML-Agents versions to use.
2. Python environment for the trainer.
3. Git branching rules and what not to upload.

3. What We Discussed

- **Unity version.** We chose Unity 2019.3.15f1 with the ML-Agents Unity package 1.0.2-preview. These versions match each other and the Hummingbirds setup we are following, so we avoid version problems at the start.
- **Python trainer.** The matching Python trainer is mlagents 0.16.1. Sir suggested a separate virtual environment with Python 3.7, so it does not clash with other Python projects on the PC.
- **Quick check.** If 'mlagents-learn --help' runs without errors, the trainer is installed correctly.
- **Git rules.** main is only for stable work, develop is for joining features, and every new piece of work goes on a feat/ branch. Changes go into develop and main only through Pull Requests.
- **What not to push.** Unity's Library and Temp folders, training results and W&B logs are big and can be re-created, so they go in .gitignore.

4. Small Implementation Part

In the meeting we went through the setup steps one by one. These are the commands we agreed on, which I then ran on the lab PC.

```
# 1) Python environment for the ML-Agents trainer (Windows)
python -m venv venv
venv\Scripts\activate
pip install mlagents==0.16.1 mlagents-envs==0.16.1
mlagents-learn --help          # works = install is fine

# 2) Git branching (gitflow)
git init
git checkout -b main
git checkout -b develop
git checkout -b feat/project-setup
```

```
# .gitignore (main parts)
[Ll]ibrary/
[Tt]emp/
[Ll]ogs/
[Bb]uild/
venv/
results/        # ML-Agents training output
wandb/          # W&B local logs
```

5. Decisions Taken

- Unity 2019.3.15f1 + ML-Agents package 1.0.2-preview + Python trainer mlagents 0.16.1.
- Python 3.7 virtual environment, packages listed in requirements.txt.
- Gitflow branches: main, develop, feat/*; Pull Request needed for main and develop.
- Big generated folders are never pushed to GitHub.

6. Action Items

#	Action Item	Responsible	Due Date
1	Finish installing Unity 2019.3.15f1 through Unity Hub.	Raza	21 Oct 2026
2	Create the empty Unity project and add the ML-Agents package.	Raza	23 Oct 2026
3	Write requirements.txt and .gitignore.	Raza	23 Oct 2026
4	Check whether these versions support RND before we reach that stage.	Raza	30 Oct 2026

7. Other Notes

- The lab PC has enough RAM, but long runs should be started in the evening so they do not block other students.

8. Next Meeting

Tuesday, 3 November 2026 - review the Unity project and the first training config draft.